BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm | Semester: Fall 2025
Duration: 1 hour 30 min | Full Marks: 40

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

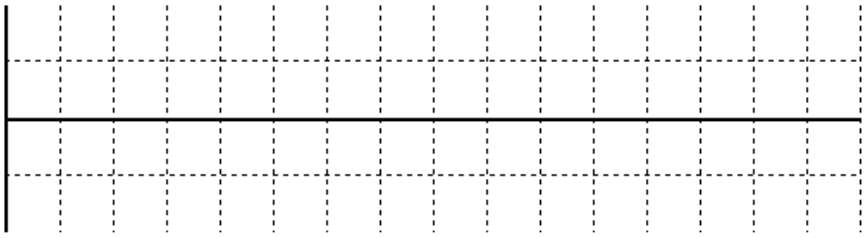
SET B

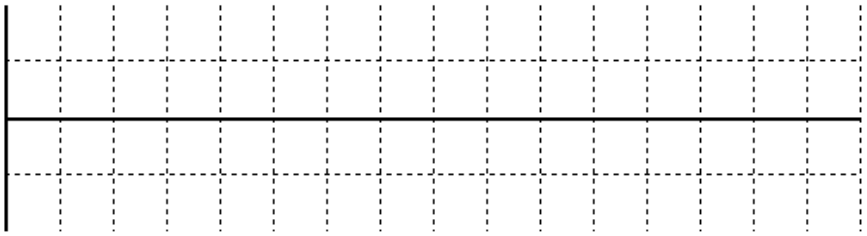
1. [CO1]	a)	Precision Manufacturing Inc. is building a new automated factory. The production process is divided into two key zones, each with unique networking demands. You are to design the **network topology** for these zones to ensure efficient and reliable operations. **Zone A:** Robotic Assembly Line This zone uses a series of 6 synchronized robotic arms. Each arm performs a specific task and must pass the product to the next arm in a precise sequence in clockwise fashion. **Zone B:** Automated Warehouse & Logistics Grid In this warehouse, 5 autonomous guided vehicles (AGVs) communicate with each other. The critical requirement is maximum reliability and redundant pathways. If one AGV breaks down or a wireless connection point fails, the others must be able to dynamically re-route their communication paths to avoid delays and maintain coordination. The two zones are connected via a central device. Your Task: Based on the scenarios and hints above, **Identify** and **design** the network topology. **Determine** the total number of links in the network.	[1+2+2]
	b)	**Identify** the layers of **TCP/IP model** which are responsible for the phrases in bold: In a smart city, a fleet of drones monitors traffic conditions and sends live updates to a central control server. Each drone continuously transmits **sensor data readings** [1] to the IoT gateway through the internet. The data is divided into smaller **segments for reliable delivery** [2] and then assigned **logical addresses** [3] so that it can be routed correctly to the control server. Before transmission, the drone's network interface organizes the data into **frames** [4] with physical addresses and sends them wirelessly to	[3]

		the gateway. Upon arrival, the gateway **reassembles the frames** [5], verifies the data, and sends an **acknowledgement of successful receipt** [6] to the drone. When a new drone joins the system, it first sends a request to identify the physical address corresponding to the gateway's network address before communication begins.	
	c)	What is Jitter and how does it affect the effectiveness of a network system? Explain why high jitter would be particularly disruptive for a live cricket match but less so for downloading an email.	[2]
	d)	Explain how process to process delivery is different from hop-to-hop delivery.	[2]
2. [CO2]	a)	A transmission line is 20 km long, where A is the source and D is the destination. To prevent excessive signal degradation, an amplifier is placed between point B and C. As a result, the received signal at point C gets amplified by 3 times than the signal received at point B. Point B is located 9 km away from the source. The signal power measured at the destination D is 15 W. The attenuation along the segment C to D is 6.80 dB. **Calculate** the signal power at point B. **[Hint. There are two segments in the overall transmission medium]**	[4]
	b)	You are transmitting a periodic composite signal to a friend who is **500 km** away through a communication channel. The signal can be decomposed into **four** sine waves with time periods of 30 ms, 15 ms, 65 ms, and 45 ms respectively. Each sine wave has a peak amplitude of 25 V. **Draw** the frequency-domain representation (spectrum) of the composite analog signal.	[2]
	c)	Mention why increasing the levels of a signal may reduce the reliability of the system?	[2]
	d)	Suppose, a document file containing 10 pages is being transmitted over a noiseless channel at a speed of 180 kbps. Each page contains 80 characters, and the file is encoded using ASCII (8 bits per character). The channel being used supports 15 signal levels. i. **Calculate** the bandwidth of the channel in Hz. If the channel is a low-pass channel, **determine** its highest frequency. ii. If the actual capacity of the channel is 13% less than the theoretical noiseless channel's capacity, **identify** the Signal-to-Noise Ratio (SNR) in decibels (dB). iii. **Compute** the transmission delay for sending the entire document file over this channel.	[3+3+1]

<table>
<tr>
<td>3.
[CO2]</td>
<td>a)</td>
<td>A measurement system records a composite analog signal that is the sum of four sinusoidal components with frequencies and amplitudes as follows: 4 Hz (7 V), 8 Hz (2 V), 15 Hz (3 V), and 10 Hz (5 V). The signal is to be digitized using a PCM system that employs 128 quantization codes (levels).

i. Determine the minimum sampling frequency required for proper construction.

ii. Identify the number of bits per sample and the resulting minimum bit rate (in bits per second) of the PCM bitstream.</td>
<td>[2+2]</td>
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<td></td>
<td>b)</td>
<td>You are designing a custom 3B/5B block encoding scheme for a digital communication system.
Each 3-bit data block will be substituted by a 5-bit codeword to improve clock recovery and limit long runs of zeros.
A valid 5-bit codeword must start with 1, must not contain ‘000’, must contain at least two 1’s and must have even number of 1’s. Four of the codewords are already provided as reference in the partial mapping table below:
<table>
<tr><th>Dataword</th><th>Codeword</th></tr>
<tr><td>000</td><td></td></tr>
<tr><td>001</td><td>11101</td></tr>
<tr><td>010</td><td></td></tr>
<tr><td>011</td><td>11110</td></tr>
<tr><td>100</td><td></td></tr>
<tr><td>101</td><td>11011</td></tr>
<tr><td>110</td><td>10111</td></tr>
<tr><td>111</td><td></td></tr>
</table>
Fill in the missing 5-bit Codewords for the remaining Dataword using valid codewords that follow all the rules given above. (Write the completed table on your question.)</td>
<td>[4]</td>
</tr>
<tr>
<td></td>
<td>c)</td>
<td>Suppose in a system, a multi-level line coding scheme will be used with the following properties:
There will be synchronization issues and a DC problem for a long sequence of 0’s, but a long sequence of 1’s does not contribute to any such problems. Which line coding technique will be followed?
Now, encode the following binary bits using the graph below:
101000100010
  </td>
<td>[2]</td>
</tr>
</table>

	d)	Apply a technique that can resolve both the synchronization issues and the DC problem for a long sequence of 0's without requiring any extra bandwidth. **What** is the name of the technique? **Encode** the following binary bits **using the graph below**: **10100000010** 	[3]

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